Windows basics: processes, services, registry, users and groups, UAC, and the 3 main Event Viewer logs (Application, Security, System)

Windows is one of the most common operating systems in enterprise environments.

A SOC analyst needs to understand how Windows works because many attacks target Windows processes, services, user accounts, authentication mechanisms, and system configuration.

1. Processes

A process is a running instance of a program.
Examples:

explorer.exe
chrome.exe
notepad.exe
powershell.exe
cmd.exe
svchost.exe

Each process has:

- Process ID (PID).
- Process name.
- Executable path.
- User account.
- Parent process.
- Memory allocation.
- Associated network connections, where applicable.

What is a parent-child process?

A parent process creates or starts another process.
Example:

```
explorer.exe
   |
   └── chrome.exe
         |
         └── powershell.exe
```

This represents a possible process chain, not necessarily malicious activity.
However, a browser spawning PowerShell unexpectedly may deserve investigation.

Suspicious process behaviour

Behaviour	Why it matters
Word spawning PowerShell	Could indicate malicious macros or document exploitation
Excel spawning cmd.exe	May indicate malicious scripting
PowerShell downloading files	Could be legitimate administration or malware delivery
Rundll32 executing unusual DLLs	Could indicate abuse of a legitimate Windows utility
Process running from Temp	Malware sometimes executes from temporary folders
Unexpected process running as SYSTEM	Could indicate privilege escalation or a service-related activity

Useful Windows commands

List processes:
Get-Process
Find a process:
Get-Process powershell
List process IDs and executable paths:
Get-CimInstance Win32_Process |
Select-Object ProcessId, ParentProcessId, Name, ExecutablePath

2. Services

A Windows service is a program designed to run in the background.
Services can start automatically, manually, or through triggers.

Examples:

- Windows Defender Antivirus Service.
- Windows Update.
- Print Spooler.
- Remote Procedure Call (RPC).

Services can run under accounts such as:

- Local System.
- Local Service.
- Network Service.
- Dedicated service accounts.

Why do attackers target services?

Attackers may:

- Create malicious services for persistence.
- Modify existing services.
- Run malicious executables as services.
- Abuse service permissions.
- Install remote access tools as services.

Example suspicious service:

Service Name: UpdateService

Executable:

C:\Users\Public\update.exe

Startup Type: Automatic

The name looks legitimate, but the executable location deserves investigation.

Useful commands

List services:

Get-Service

Check a specific service:

Get-Service -Name Spooler

Find service configuration:

sc qc Spooler

3. Windows Registry

The Windows Registry is a hierarchical database that stores configuration information for Windows, applications, users, and hardware.

It contains keys, subkeys, and values.

Important registry hives

Hive	Purpose
HKLM	HKEY_LOCAL_MACHINE: System-wide settings
HKCU	HKEY_CURRENT_USER: Current user's settings
HKCR	HKEY_CLASSES_ROOT: File associations and COM information
HKU	HKEY_USERS: User profiles
HKCC	HKEY_CURRENT_CONFIG: Current hardware profile

Important registry locations for SOC investigations

a. Run keys

HKCU\Software\Microsoft\Windows\CurrentVersion\Run

HKLM\Software\Microsoft\Windows\CurrentVersion\Run

Why important?

Malware may add an entry to these locations to execute automatically when a user logs in.

b. Services

HKLM\SYSTEM\CurrentControlSet\Services

Why important?

Malicious or unauthorized services may be registered here.

c. Winlogon

HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Winlogon

Why important?

Changes to logon-related settings can be used for persistence.

d. Windows Defender settings

HKLM\SOFTWARE\Microsoft\Windows Defender

Why important?

Unexpected security configuration changes may indicate attempted defence evasion.

Access and visibility depend on Windows version and security protections.

Suspicious registry behaviour

- New Run key entries.
- Unexpected service registrations.
- Changes to security settings.
- Modifications to logon-related values.
- Unexpected startup locations.

4. Windows users and groups

A user account represents an identity that can authenticate to a Windows system.

Groups are used to manage permissions for multiple users.

Common Windows groups

Group	Purpose
Administrators	Full administrative control
Users	Standard user permissions
Remote Desktop Users	Allowed to log in through Remote Desktop
Backup Operators	Special backup and restore privileges
Guests	Restricted guest access
Event Log Readers	Can read event logs

Why are users and groups important to SOC?

Attackers may:

- Create unauthorized accounts.
- Add themselves to Administrators.
- Enable disabled accounts.
- Modify account permissions.
- Abuse privileged accounts.
- Create accounts for persistence.

Example:

An employee's account suddenly gets added to the local Administrators group.

This may indicate:

- Authorized IT administration.

- Privilege escalation.
- Compromised account activity.

A SOC analyst should investigate who performed the action and whether it was approved.

Useful commands

List local users:
Get-LocalUser
List local groups:
Get-LocalGroup
List local administrators:
Get-LocalGroupMember Administrators

5. UAC (User Account Control)

UAC stands for User Account Control.

It is a Windows security feature that helps prevent unauthorized administrative changes.
When an application requests administrative privileges, Windows may display a consent prompt.

Example:
A standard user attempts to install software requiring administrative permissions.
Windows prompts for an administrator's credentials.

UAC concepts

Concept	Meaning
Standard user	Has ordinary permissions
Administrator account	Has administrative capabilities
UAC prompt	Requests consent or credentials for elevation
Elevation	Running a process with higher privileges
Integrity level	Security level associated with a process

Common integrity levels include:

- Low.
- Medium.
- High.
- System.

Why does UAC matter to a SOC analyst?

Attackers may attempt to:

- Bypass UAC.
- Elevate privileges.
- Execute processes with administrator privileges.
- Modify system configuration.

Suspicious example:
A user opens a document, which launches a script that attempts to execute an administrative process unexpectedly.

An analyst should inspect the process tree, user context, UAC-related events, and subsequent activity.

Event Viewer
Event Viewer is a Windows tool used to inspect operating system and application events.
It is extremely important for SOC investigations.
The three main logs are:

1. Application.
2. Security.
3. System.

1. Application log
Contains events generated by applications and software.

Examples:

- Application crashes.
- Software errors.
- Application startup issues.
- Database errors.
- Application-specific warnings.

SOC use case:
If an application unexpectedly crashes or an unusual program produces repeated errors, the Application log may help explain what happened.

2. Security log
Contains security-related events generated by Windows auditing.

Examples:

- Successful logons.
- Failed logons.
- Account creation.
- Account modification.
- Privilege assignment.
- Process creation, when auditing is enabled.
- Security log clearing.

This is one of the most important logs for SOC analysts.

SOC use case:

Investigate brute-force attacks, suspicious logons, unauthorized account creation, and privilege escalation.

3. System log

Contains events generated by Windows system components.

Examples:

- Service startup and shutdown.
- Driver problems.
- Unexpected shutdowns.
- Hardware-related issues.
- System component failures.

SOC use case:
Investigate suspicious service installation, service failure, unexpected reboots, and system-level problems.

4. Comparison

Log	What it records	SOC investigation example
Application	Application-related events	Suspicious application crash
Security	Authentication and security auditing	Brute-force attack
System	Windows system components	Unauthorized service activity

Remember: Event IDs are not guaranteed to be generated on every machine. Audit policies, Windows versions, configuration, permissions, and event forwarding determine what is available.